

Amazon E-Commerce Intelligence Dashboard Using Power BI

1. Introduction

The Amazon E-Commerce Intelligence Dashboard is an interactive business intelligence solution developed using Power BI. The dashboard provides insights into sales performance, customer satisfaction, category performance, and operational efficiency. It helps stakeholders monitor business performance and make data-driven decisions through interactive visualizations and filters.

2. Objectives

- Analyze sales and revenue performance.
- Monitor order trends across different years.
- Evaluate product category performance.
- Track customer satisfaction and return rates.
- Provide interactive analysis through slicers and filters.

3. Tools and Technologies Used

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Modeling
- Interactive Visualizations
- Slicers and Filters

4. Dataset Description

The dataset contains Amazon e-commerce transaction records including:

- Order ID
- Order Date
- Product Category
- Price
- Discount
- Customer Rating
- Delivery Status

5. Data Preparation and Transformation

The following preprocessing steps were performed:

- Removed duplicate records.
- Checked and handled missing values.
- Corrected data types.
- Created Month-Year column.
- Created Month Sort column to display months in chronological order.
- Loaded transformed data into Power BI.

Date Transformation

A Month Sort column was created to ensure that months are displayed from January to December rather than alphabetically. This improved the accuracy of monthly trend analysis.

6. KPI Metrics

The dashboard includes:

- **Total Revenue** – Measures overall sales performance.
- **Total Orders** – Tracks customer demand and order volume.
- **Average Order Value** – Analyzes average customer spending per order.
- **Return Rate** – Evaluates the percentage of returned orders.
- **Delivery Success Rate** – Monitors delivery efficiency.
- **Customer Satisfaction Rating** – Measures customer experience and product satisfaction.



7. Year-Wise Analysis



The company generated ₹4 billion in revenue during 2024. Electronics was the leading category, while customer satisfaction remained stable. Revenue trends were consistent throughout the year.



Revenue increased by approximately 25% compared to 2024. Order volume also increased significantly, indicating business growth. Electronics continued to dominate category sales.



Revenue remained stable during the first quarter of 2026. Return rates improved slightly, and Electronics remained the strongest category.

8. Category-Wise Analysis

Electronics



Electronics is the highest-performing category and the primary contributor to overall revenue. It achieved the highest customer satisfaction score and the lowest return rate.

Home



Home products are the second-largest revenue contributor and demonstrate stable performance across all metrics.

Sports



Sports products maintain strong order volume but generate lower revenue due to lower average order values

Beauty



The Beauty category generates lower revenue despite maintaining a strong order volume. Improvements in customer engagement and product offerings may increase performance.

Clothing



The Clothing category has the lowest average order value and revenue contribution. Strategic pricing and promotional activities can help improve category growth.

Slicers and Filters

The dashboard includes interactive slicers that allow users to dynamically analyze data from different perspectives.

- **Year Filter** – Enables year-wise performance analysis and comparison.
- **Quarter Filter** – Allows users to analyze business performance across different quarters and identify seasonal trends.
- **Category Filter** – Allows users to evaluate performance across different product categories.
- **Delivery Status Filter** – Helps analyze order fulfillment and delivery performance.
- **Customer Rating Filter** – Enables assessment of customer satisfaction and product quality.

10. Key Insights

- Revenue increased from ₹4 billion in 2024 to ₹5 billion in 2025.
- Electronics is the highest-performing category with ₹7 billion revenue.
- Electronics achieved the highest customer satisfaction rating (4.21).
- Home and Sports contribute significantly to overall sales.
- Beauty and Clothing generate the lowest revenue despite similar order volumes.
- Return rates remain relatively stable across categories.

11. Business Recommendations

- Continue investing in the Electronics category.
- Increase promotional activities for Home and Sports categories.
- Improve product quality and descriptions to reduce return rates.
- Introduce premium products and bundle offers in lower-performing categories.
- Monitor delivery performance to enhance customer satisfaction.

12. Conclusion

The Amazon E-Commerce Intelligence Dashboard successfully provides a comprehensive view of sales performance, customer behavior, category trends, and operational metrics. Through KPI monitoring, interactive visualizations, and slicers, the dashboard supports data-driven decision-making and helps identify opportunities for business growth.